



Basic Details of the Team and Problem Statement

Ministry/Organization Name/Student Innovation:

All India Council for Technical Education(AICTE) & Ministry of Innovation's Council(MIC)

PS Code: SIH1498

Problem Statement Title: Develop a Proctored Exam Tool for shortlisting the candidates for national and international level hackathons

Team Name: The Googlers

Team Leader Name: Het Patel

Institute Code (AISHE): C-94

Institute Name: G H Patel College of Engineering and Technology

Theme Name: Smart Automation

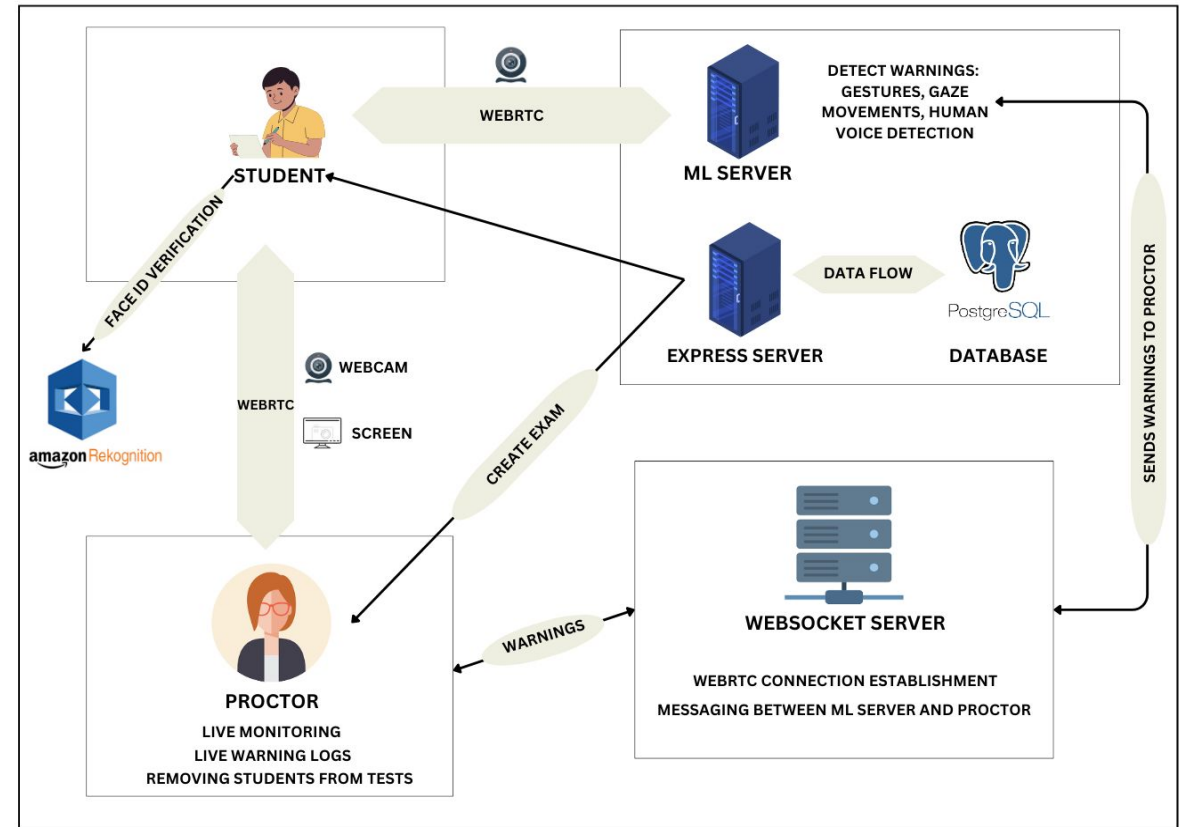
Idea/Approach Details

Solution/Prototype:

Vision: Empowering Academic Integrity and Learning in a Digital Age

Mission: We exist to ensure academic integrity, fairness, and student success in online education through a secure and transparent examination environment.

- ❖ Our proposed solution consists 1 website for **proctor** and 1 desktop app for **students**.
- ❖ Proctor can create exam and can see **facecam** and **screen recording** of each student who is giving exam.
- ❖ Students have to give exam on our **browser app** which will restrict students from doing any suspicious activity.
- ❖ Live video frames of students will be sent to **CV model** which will do facial recognition, eye tracking, mobile detection and multiple faces detection.
- ❖ **Audio** of student will be captured and processed for human voice detection.



Technology Stack :

- Frontend: ReactJS
- Browser: ElectronJS
- Backend: NodeJS , ExpressJS , WebRTC
- Database: PostgreSQL
- CV: Yolo
- AWS: EC2, S3, Rekognition, RDS

Idea/Approach Details

Use Cases:



Stoppers:

- ❑ **Our Own Browser:** On website, we can't control hardware components like ctrl+c, ctrl+v, alt+tab, windows, ports, finger gesture and screen sharing apps, but by having our own browser we can solve this issue.
- ❑ **Live monitoring:** We can see webcam and screen of all the candidates/employees.
- ❑ **Reducing human efforts:** To monitor each and every student's webcam and screen is very tough process, we have automated this monitoring by applying AI.

Dependencies:

- ❑ Windows operating system.
- ❑ Stable internet connection.

Team Member Details

Team Leader Name: Het Patel

Branch : B.E. Stream : IT Year : IV

Team Member 1 Name: Pratham Shah

Branch : B.E. Stream : IT Year : IV

Team Member 2 Name: Ayushi Patel

Branch : B.E. Stream : IT Year : IV

Team Member 3 Name: Kandarp Shah

Branch : B.E. Stream : CP Year : IV

Team Member 4 Name: Nakshatra Vyas

Branch : BTech Stream : CP Year : III

Team Member 5 Name: Krish Jotaniya

Branch : BTech Stream : CSE(IOT) Year : III

Team Mentor 1 Name: Hetal Gaudani

Category : Academic Expertise : AI/ML Domain Experience : 15 years